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DeepMark Benchmark: Redefining Audio Watermarking Robustness

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Abstract

Evaluating audio watermarking algorithms against modern generative attacks requires standardized benchmarks. DeepMark provides a comprehensive standardized testbed and open dataset for benchmarking traditional DSP watermarking (DWT, SVD, LSB, Patchwork) against deep learning neural watermarkers under realistic transmission attacks, MP3/AAC compression, pitch shifting, time stretching, and generative AI deepfake voice synthesis.

Key Methodology & Architecture

Standardized evaluation pipeline, end-to-end differentiable noise layers, robustness score matrix, perceptual evaluation of audio quality (PEAQ).

Datasets & Benchmark Environments

DeepMark Benchmark Suite (Common Voice, VCTK, GTZAN music dataset).

Performance Metrics & Graph Axis Parameters

Reported Results: Robustness Score (0-100), PEAQ ODG (-0.2 to -4.0), BER under AAC 64 kbps < 1.2%.

Graph Parameter Mapping: X-axis: Attack Type (MP3 64k, Additive Noise, Pitch Shift, AI Voice Synth); Y-axis: Watermark Recovery Rate (0% to 100%).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Comprehensive comparison across traditional and deep learning watermarking paradigms.	Benchmark dataset size requires substantial disk storage for full evaluation.

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